MILITARY SPECIFICATION

DISPENSER, CARBONATED BEVERAGE, MECHANICALLY REFRIGERATED; COUNTERTOP DISPENSING, REMOTE SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanically refrigerated post-mix, pressure type carbonated beverage dispensers with remote countertop dispensing units.

1.2 Classification. Beverage dispensers shall be of the following sizes as specified (see 6.2).

Size 1 - 300 six-ounce drinks/hr capacity
Size 2 - 480 six-ounce drinks/hr capacity
Size 3 - 900 six-ounce drinks/hr capacity

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated Triple-Wall

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute (ANSI)

C73 - Dimensions of Caps, Plugs, and Receptacles

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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National Sanitation Foundation (NSF)

Standard No. 18 - Manual Food and Beverage Dispensing Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

Underwriters' Laboratories, Inc., (UL)

Standard No. 471 - Commercial Refrigerators

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio St., Chicago, IL 60611; 1285 Walt Whitman Road, Melville, Long Island, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. Dispensers delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.2 Codes and standards. The dispensers shall comply with the applicable requirements of NSF Standard No. 18 and UL Standard No. 471.

3.2.1 Compliance. Prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the dispensers furnished under this specification meet the applicable requirements of UL and NSF as follows:

3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished dispenser, or

(2) A certification for the dispensers issued by NSF under their special one-time contract evaluation/certification service, or

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(3) A certified test report from a recognized independent testing laboratory acceptable to the Army Medical Department indicating that the dispensers have been tested and conform to NSF Standard No. 18.

3.2.1.2 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the dispensers have been tested and conform to UL Standard No. 471.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial dispenser provided the completed item complies with all provisions of this specification.

3.4 Design and construction. The dispenser shall be a mechanically refrigerated, post-mix, pressure flow type, consisting of a floor mounted refrigeration unit, counter mounted dispensing heads, and insulated conduit for remote dispensing head installation.

3.4.1 Refrigeration unit. The refrigeration unit shall consist of a compressor, condenser, pump, ice bank (when used), carbonator, electrical and plumbing connections. Maximum allowable overall dimensions shall be: length 33 inches, width 26 inches, and height 32 inches. The refrigeration system shall be hermetically sealed and shall continuously maintain drink temperatures under 42F at the dispensing head. The carbonator shall be capable of supplying finished drink carbonation levels at a minimum of 3 gas volumes with inlet water ranging from 50F to 90F. Unless otherwise specified (see 6.2), the unit shall operate on water pressure between 30 and 80 pounds per square inch gage (psig). Easily accessible connections with quick disconnect fittings shall be provided for cold water and CO_2 gas supply. Each post-mix valve shall give a constant rate of flow, controlling the water and syrup to provide the syrup manufacturer's recommended Brix using syrup temperatures from 40 to 90F. Brix stratification in the finished drink shall be $\pm$ 0.5 of the syrup manufacturer's recommended Brix from top to bottom, when tested as specified in 4.4.1. Syrup flow shall be adjustable to obtain the desired Brix. Each valve shall be easily removable with normal hand tools for servicing without interrupting the operation of the other dispensing valves. When specified (see 6.2), a water filter shall be furnished. The filter shall be the replaceable cartridge type, constructed of corrosion-resistant metals and shall be capable of removing particles of turbidity larger than 2 microns in size and 3.0 parts per million (ppm) chlorine from the influent water.

3.4.1.1 Electrical requirements. Unless otherwise specified (see 6.2), the dispenser shall operate on a 120 volt, single phase, 60 hertz, electrical system. When 50 hertz operation is specified (see 6.2), a UL approved, 3-wire power supply cord without an attachment plug shall be provided, except when specified

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(see 6.2), the supply cord shall be provided with an attachment plug in accordance with ANSI C73.

3.4.2 Dispensing heads. A dispensing head shall have 4 spigots, a drip pan and a drain assembly. Unless otherwise specified (see 6.2), valves shall be electrically actuated, operating on a 24 volt AC system. Each dispenser shall be provided with one 115 volt to 24 volt AC system. Each power the dispensing heads. Maximum allowable dispensing head dimensions shall be: width 14 inches, depth 12 inches, and height 16 inches. One or two dispensing heads as specified (see 6.2), shall be included with each dispenser.

3.4.3 Insulated conduit. Insulated conduit with 4 syrup lines, in the length specified (see 6.2), shall be supplied with each dispenser. Necessary connections and fittings to hook up the specified number of dispensing heads (see 6.2), shall also be supplied.

3.4.4 Performance. Under conditions of 75F ambient, 75F syrup and 75F inlet water, the dispensers shall provide drinks at capacities not less than those shown in table I, when tested as specified in 4.4.1. Dispensed drinks shall have a temperature under 42F, a minimum of 3 gas volumes carbonation, and a Brix within 0.5 of that recommended by the syrup manufacturer.

TABLE I. Capacity requirements

Dispenser size	Dispensing capacity (6-oz drinks/hr)	Recovery capacity (6-oz drinks/hr)
1	300	120
2	480	180
3	900	360

3.5 Marking. Each dispenser shall be permanently and legibly marked with the manufacturer's model and name, trade name, or trademark easily identified with said manufacturer.

3.6 Workmanship. The dispenser shall be free of any burrs, slivers, sharp edges, or sharp corners which may cause injury or affect the service life of any component. Welds shall not contain cracks, burn holes, or have incomplete fusion and all scale and flux shall be removed. Threaded fasteners shall not be missing, stripped or loose; rivets shall not be loose or not in full contact with members riveted.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall be all dispensers of one size offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated dispenser.

4.2.2.1 Visual examination. The dispensers shall be examined for the defects in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturers standard product	X	
Refrigeration unit	Quick disconnect fittings not supplied on cold water and CO_2 connections		X
	Post-mix valves not as specified	X	
	Filter (when supplied) not as specified		X

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Dispensing heads	Number specified not provided	X	
	Drain missing, drip pan missing		X
	Valves not type specified		X
	Transformer missing (electric valves only) or specified type not provided	X	
Conduits	Missing, not as specified	X	
	Fittings not provided to hook up dispensing heads	X	
Workmanship			
Finish	Burrs, slivers, sharp edges, sharp corners which may cause injury or affect service life of any component	X	
Welds	Cracks, incomplete fusion, burn holes	X	
	Scale and flux not removed		X
Threaded fasteners	Stripped, loose		X
	Missing	X	
Rivets	Loose, not in full contact with members riveted		X
Marking	Missing, incomplete, illegible		X

4.2.2.2 Dimensional examination. The dispensers shall be examined for compliance with dimensions specified. Any dimensions not within the specified requirements shall be a defect. The inspection level shall be S-4 with an AQL of 6.5 expressed in defects per hundred units.

4.2.3 End item testing. Each sample dispenser shall be tested as specified in 4.4.1; failure of this test shall be recorded as a defect. The inspection level shall be S-4 with an AQL of 4.0 per hundred units.

4.2.4 Examination of preparation for delivery. An examination shall be made to determine that packing and marking as required by section 5 are complied with.

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Defects shall be as indicated in table III. The sample unit shall be one shipping container offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE III. Examination of preparation for delivery

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.

4.2.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-640, examination for defects in closure and strapping shall be in accordance with the applicable container specification. Examination for defects in cushioning, blocking and bracing, and waterproofing shall be in accordance with MIL-STD-1186.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Tests. Prior to running the following test the manufacturer shall operate the dispenser for a period of time to verify that the components and controls are functioning properly.

4.4.1 Operational test. The test setup for size 1 dispensers shall be one dispensing head connected to the refrigeration unit by approximately 30 feet of insulated conduit. Size 2 and 3 dispensers shall have two dispensing heads connected to the conduit; one approximately 30 feet and one approximately 60 feet from the refrigeration unit. With a tolerance of $\pm$ 2%, the ambient temperature, syrup temperature and inlet water temperature all shall be 75F. With appropriate carbonate pressure and water pressure within the manufacturer's recommended limits, the dispensers shall be operated for a period of 60 minutes to determine conformance with the capacity specified in table I. Size 2 and 3 dispensers shall have approximately equal amounts of drink dispensed from each dispensing head. The first, middle, and last drinks dispensed shall have a measured temperature under 42F, a minimum of 3 gas volumes carbonation, and a

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Brix within ± 0.5 of that recommended by the syrup manufacturer. Once set, the Brix control regulator shall not be changed during this entire operational test. The Brix shall be determined as the average at the top and bottom of each drink with a refractometer equal to American Optical or Bauch and Lomb types; the carbonation shall be determined by a Bastian and Blessing carbonation tester or equal. Immediately at the end of the hour, the dispensers shall be run under the same conditions for an additional period of 60 minutes to determine conformance with the recovery capacity specified in 3.4.4. The middle and last drinks dispensed shall have a measured temperature under 42F, a minimum of 3 gas volumes, and a Brix within ± 0.5 of that recommended by the syrup manufacturer. The Brix and carbonation shall be determined as specified above. Inability of the dispenser to meet the requirements specified in 3.4.1 and 3.4.4 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each dispenser shall be packed in a cleated-plywood or nailed wood shipping container conforming to overseas type, style A or B, grade A or B, type 2 load of PPP-B-601 or class 2, style 2 or 4, grade A or B, type 2 load of PPP-B-621, respectively. Cushioning, blocking and bracing, and waterproofing of the contents of each shipping container shall be in accordance with MIL-STD-1186. Each shipping container shall be closed and strapped in accordance with the applicable container specification.

5.1.2 Level B. Each dispenser shall be packed in a shipping container conforming to grade VIIc, style optional of PPP-B-636, or class weather-resistance, style optional of PPP-B-640, or domestic type, style A or B, grade B, type 2 load of PPP-B-601. Components of each dispenser shall be blocked, braced and cushioned in accordance with MIL-STD-1186. Each shipping container shall be closed in accordance with the appendix of the applicable container specification.

5.1.3 Level C. The dispensers shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dispensers are intended for use in feeding areas to dispense carbonated drinks. Dispensing towers are used in the serving line while the refrigeration system is remotely located.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size dispenser required (see 1.2).
(c) Water supply pressure if other than 30-80 psig (see 3.4.1).
(d) Whether a water filter is required (see 3.4.1.)
e) Electrical characteristics if other than specified (see 3.4.1.1).
(f) When 50 hertz electrical operation is required, and whether an attachment plug is required (see 3.4.1.1).
(g) When manual dispensing head valves are required (see 3.4.2).
(h) Number of 4 spigot dispensing heads required per dispenser (see 3.4.2 and 3.4.3).
(i) Length of insulated conduit required per dispenser (see 3.4.3).
(j) Selection of the applicable level of packing (see 5.1).

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DOCUMENT IDENTIFIER AND TITLE MIL-D-43913(GL) - DISPENSER, CARBONATED BEVERAGE, MECHANICALLY REFRIGERATED; COUNTERTOP DISPENSING, REMOTE SYSTEM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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